

Concentric Zero-Spheres of a class of sections of the commutative ring $\mathcal{R}$ of slice preserving functions over the compactified Octonions with the Riemann Hypothesis as a corollary

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“...a ‘viewpoint’ by itself remains fragmentary. It reveals to us one of the aspects of a scenery or panorama, among a multiplicity of others which are equally valuable, equally ‘real’. It is when complementary viewpoints of a common reality are conjugated, that is, when our ‘eyes’ are multiplied, that the gaze is able to penetrate further ahead in the reckoning of things. ...it also happens, sometimes, that a sheaf of viewpoints converging to a unique and vast scenery gives rise to a novel thing; a thing which transcends each of the partial perspectives, in the same way that a living being transcends each of its limbs and organs. This new thing could be called a vision.”

— A. Grothendieck, *Récoltes et Semailles*

Abstract

Let $\mathcal{R}$ be the commutative ring of slice-preserving slice-regular functions on the compactified octonions $\mathbb{O}^* = S^8$, under the regular $(*)$ product. We prove the *Concentricity Theorem* for a section $A \in \mathcal{R}$ satisfying four conditions: (C1) meromorphic continuation with a single simple pole carried to $\infty = N$; (C2) an infinite Euler product; (C3) an infinite slice-regular Weierstrass factorization, whose spherical factors are the residue- $\mathbb{C}$ zeros; and (C4) infinitely many residue- $\mathbb{C}$ zeros. From the action groupoid $\mathcal{H}_1 = G_2 \ltimes \mathbb{O}^*$, whose non-real orbits are 6-spheres, the slice world $\mathcal{S}_2$ of slice Riemann spheres with the band $U(1)$, and the section functor $\Phi : \mathcal{H}_1 \rightarrow \mathcal{S}_2$, we form the base $\mathcal{B}$ of the exponential’s degenerate set $\mathcal{B} \subset \mathcal{S}_2$ and the Grothendieck construction $\mathcal{T}_A = \int_{\mathcal{B}} F$ of the band $F = U(1)$ over $\mathcal{B}$, whose cocartesian transport is the winding lift. The theorem is that the residue- $\mathbb{C}$ zero 6-spheres of A lie in a single connected component of $\mathcal{T}_A$, arising as the degenerate fibre of its transport. The conclusion is computed at π_0 by cocartesian diagram chasing, through the identity $\pi_0(\mathcal{T}_A) \cong \text{colim}_{\mathcal{B}}(\pi_0 \circ F)$; the analytic slice-preserving inputs enter as cited statements. As downstream corollaries, the nontrivial-zero 6-spheres of the octonionic zeta function $\zeta_{\mathbb{O}^*}$ are concentric about one real centre c ; hence every classical nontrivial zero has real part c ; and the functional equation forces $c = \frac{1}{2}$, giving the Riemann Hypothesis. Nothing in this document has yet been formalized in Lean.

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Part I

Classical Background

1 The Riemann zeta function

Theorem 1.1 (Meromorphic continuation and functional equation; Riemann [13]). *The Dirichlet series $\zeta(s) = \sum_{n \geq 1} n^{-s}$, convergent for $\operatorname{Re} s > 1$, continues to a meromorphic function on $\mathbb{C}$ with a single simple pole, at $s = 1$; equivalently, a holomorphic map to the Riemann sphere*

$$\zeta : \mathbb{C}^* \longrightarrow \mathbb{C}^*, \quad \mathbb{C}^* = \mathbb{C} \cup \{\infty\} = S^2, \quad \zeta(1) = \infty.$$

The completed function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is entire and satisfies

$$\xi(s) = \xi(1-s), \quad \xi(\bar{s}) = \overline{\xi(s)}.$$

Consequently, if ρ is a nontrivial zero of ζ then so are $\bar{\rho}$, $1-\rho$, and $1-\bar{\rho}$. The trivial zeros of ζ are at $s = -2, -4, \dots$; the nontrivial zeros lie in the strip $0 < \operatorname{Re} s < 1$. The Riemann Hypothesis asserts that every nontrivial zero has $\operatorname{Re} s = \frac{1}{2}$.

Theorem 1.2 (Infinite Euler product; [15, Ch. 1]). *For $\operatorname{Re} s > 1$,*

$$\zeta(s) = \prod_{p \text{ prime}} (1 - p^{-s})^{-1} = \exp\left(\sum_p \sum_{k \geq 1} \frac{1}{k} p^{-ks}\right),$$

the product taken over the infinitely many primes; it converges absolutely and is zero-free on $\operatorname{Re} s > 1$.

Theorem 1.3 (Infinite Hadamard product; [15, Ch. 2]). ξ is entire of order 1 with infinitely many zeros, which are exactly the nontrivial zeros $\{\rho\}$ of ζ , and admits the Hadamard factorization

$$\xi(s) = \xi(0) \prod_{\rho} \left(1 - \frac{s}{\rho}\right) e^{s/\rho},$$

the product taken over the infinitely many nontrivial zeros.

Corollary 1.4 (Infinitude of the nontrivial zeros). ζ has infinitely many nontrivial zeros.

Proof. Immediate from Theorem 1.3: its zeros are infinitely many and are exactly the nontrivial zeros of ζ . $\square$

Theorem 1.5 (Hardy [11]). Infinitely many nontrivial zeros of ζ lie on the line $\operatorname{Re} s = \frac{1}{2}$.

The compactified Riemann zeta on $\mathbb{C}^*$

For the slice construction, ζ is taken in compactified form, with values on the Riemann sphere $\mathbb{C}^* = \mathbb{C} \cup \{\infty\} = S^2$ (the one-point compactification of $\mathbb{C}$ [12]). The classical facts above are retained as stated.

Definition 1.6 (Compactified classical zeta). $\zeta_{\mathbb{C}^*} : \mathbb{C}^* \rightarrow \mathbb{C}^*$ is the meromorphic continuation of ζ regarded as a map to the Riemann sphere: $\zeta_{\mathbb{C}^*}(s) = \zeta(s)$ for $s \in \mathbb{C} \setminus \{1\}$; $\zeta_{\mathbb{C}^*}(1) = \infty$ (the simple pole, Theorem 1.1); and $\zeta_{\mathbb{C}^*}(\infty) = 1$, since $\zeta(s) \rightarrow 1$ as $\operatorname{Re}(s) \rightarrow +\infty$ (the p -test: only the $n = 1$ term of $\sum_n n^{-s}$ survives). It is holomorphic into $\mathbb{C}^*$ on $\mathbb{C}^* \setminus \{1\}$, its sole pole being $s = 1$. The functional equation, the conjugate symmetry $\zeta_{\mathbb{C}^*}(\bar{s}) = \overline{\zeta_{\mathbb{C}^*}(s)}$, and Hardy's infinitude (Theorem 1.5) hold verbatim for $\zeta_{\mathbb{C}^*}$, as they concern values on $\mathbb{C} \setminus \{1\}$ where $\zeta_{\mathbb{C}^*} = \zeta$.

Lemma 1.7 (Compactification does not move the zeros). $Z(\zeta_{\mathbb{C}^*}) = Z(\zeta)$: the zeros of $\zeta_{\mathbb{C}^*}$ in $\mathbb{C}^*$ are exactly the zeros of ζ in $\mathbb{C}$; in particular the nontrivial zeros coincide.

Proof. On $\mathbb{C} \setminus \{1\}$ the two functions agree, so they share all zeros there. The two adjoined values are not zeros: $\zeta_{\mathbb{C}^*}(1) = \infty$ (a pole) and $\zeta_{\mathbb{C}^*}(\infty) = 1$; and $s = 1$, being a pole of ζ , was never a zero. Hence no zero is gained or lost. $\square$

Part II

Slice-Preserving Theory, the Ring $\mathcal{R}$, and the Equivalence

2 Octonions and slices

The octonions $\mathbb{O}$ form an 8-dimensional real algebra with basis $\{1, e_1, \dots, e_7\}$, each $e_i^2 = -1$.

Definition 2.1. The octonions are: (i) a *division algebra*: every nonzero element has a multiplicative inverse; (ii) a *normed algebra*: $|ab| = |a||b|$; (iii) *non-associative*: $(ab)c \neq a(bc)$ in general; (iv) *alternative*: $(aa)b = a(ab)$ and $a(bb) = (ab)b$.

An octonion $s \in \mathbb{O}$ decomposes as $s = \sigma + w$ with $\sigma = \operatorname{Re}(s) \in \mathbb{R}$ and $w = \operatorname{Im}(s) \in \operatorname{Im}(\mathbb{O}) \cong \mathbb{R}^7$; the unit imaginary octonions form $S^6 \subset \operatorname{Im}(\mathbb{O})$.

Theorem 2.2 (Artin [14]). *In an alternative algebra, the subalgebra generated by any two elements is associative.*

Corollary 2.3. *Powers s^n are well-defined for any $s \in \mathbb{O}$.*

Definition 2.4 (Complex slices and slice Riemann spheres). For any unit imaginary octonion $v \in S^6$, the *complex slice* is $\mathbb{C}_v = \{a + bv : a, b \in \mathbb{R}\} \subset \mathbb{O}$. Since $v^2 = -1$, this is a subalgebra isomorphic to $\mathbb{C}$ via the $\mathbb{R}$ -algebra isomorphism $\varphi_v : \mathbb{C} \rightarrow \mathbb{C}_v, i \mapsto v$. All slices share the real axis: $\mathbb{C}_v \cap \mathbb{C}_w = \mathbb{R}$ for $v \neq \pm w$. Its one-point compactification is the *slice Riemann sphere* $\mathbb{C}_v^* := \mathbb{C}_v \cup \{\infty\} = S_v^2$, and φ_v extends to an isomorphism of Riemann spheres $\varphi_v : \mathbb{C}^* \xrightarrow{\sim} \mathbb{C}_v^*$ with $\varphi_v(\infty) = \infty$ (GPV [17], Prop. 4.1/Thm. 4.2). All slice spheres share the single point at infinity, as well as the real axis.

3 Slice-preserving functions

The class we use is the *slice-preserving* slice-regular functions. We first recall slice regularity [8, 6, 5, 9] — holomorphicity on each complex slice separately — as the ambient notion, then isolate the slice-preserving subclass, on which classical complex methods apply slicewise (by Artin’s theorem each slice $\mathbb{C}_v$ is an associative subalgebra where classical complex analysis applies unchanged).

Definition 3.1 (Slice regularity; [5, Def. 5.1.1]). Let $\Omega \subseteq \mathbb{O}$ be open. A function $f : \Omega \rightarrow \mathbb{O}$ is *slice regular* if for each $v \in S^6$, $\varphi_v^{-1} \circ f \circ \varphi_v$ satisfies the Cauchy–Riemann equations on $\varphi_v^{-1}(\Omega \cap \mathbb{C}_v) \subseteq \mathbb{C}$. A domain Ω is *axially symmetric* if $\sigma + \gamma v \in \Omega$ implies $\sigma + \gamma w \in \Omega$ for all $w \in S^6$.

Theorem 3.2 (Representation Formula; [5, Thm. 5.1.7]). *Let f be slice regular on an axially symmetric slice domain Ω . Fix $v_0 \in S^6$ and write $f(\sigma + \gamma v_0) = \alpha(\sigma, \gamma) + v_0 \cdot \beta(\sigma, \gamma)$ with $\alpha, \beta : \mathbb{R}^2 \rightarrow \mathbb{R}$. Then for every $v \in S^6$, $f(\sigma + \gamma v) = \alpha(\sigma, \gamma) + v \cdot \beta(\sigma, \gamma)$, where $\alpha = \frac{1}{2}[f(\sigma + \gamma v_0) + f(\sigma - \gamma v_0)]$ and $\beta = \frac{1}{2}v_0^{-1}[f(\sigma - \gamma v_0) - f(\sigma + \gamma v_0)]$.*

Theorem 3.3 (Identity Theorem; [5, Cor. 5.1.9]). *Let f be slice regular on an axially symmetric slice domain Ω . If f vanishes on a subset of a single slice $\Omega \cap \mathbb{C}_{v_0}$ with an accumulation point in $\Omega \cap \mathbb{C}_{v_0}$, then $f \equiv 0$ on all of Ω .*

Theorem 3.4 (Extension Theorem; [5, Thm. 5.1.5]). *Let $U \subseteq \mathbb{C}$ be a domain symmetric under conjugation and $g : U \rightarrow \mathbb{C}$ holomorphic. There is a unique slice regular $\text{ext}(g) : \Omega_U \rightarrow \mathbb{O}$ on $\Omega_U = \{\sigma + \gamma v : \sigma + i\gamma \in U, v \in S^6\}$ with $\text{ext}(g)|_{\mathbb{C}_{v_0}} = \varphi_{v_0} \circ g \circ \varphi_{v_0}^{-1}$ for every v_0 .*

Definition 3.5 (Slice-preserving functions; [1, Def. 2.7, Rem. 2.8]; [17, 7, 16]). A slice-regular function f on an axially symmetric domain Ω is *slice preserving* if it maps every slice into itself: $f(\Omega \cap \mathbb{C}_v) \subseteq \mathbb{C}_v$ for all $v \in S^6$. For octonions ($\#S^6 > 2$) the following are equivalent characterisations:

- (i) f is *intrinsic*: $f(q^c) = f(q)^c$ for all $q \in \Omega$, where q^c is octonionic conjugation (on a slice, $a + bv \mapsto a - bv$);
- (ii) its representation-formula components are $\mathbb{R}$ -valued: writing $f(\sigma + \gamma v) = \alpha(\sigma, \gamma) + v \beta(\sigma, \gamma)$ (Theorem 3.2), one has $\alpha, \beta : \mathbb{R}^2 \rightarrow \mathbb{R}$ — equivalently the inducing stem function $F = F_1 + \iota F_2$ has $\mathbb{R}$ -valued components F_1, F_2 ;
- (iii) f preserves two distinct slices; equivalently $f(\Omega \cap \mathbb{R}) \subseteq \mathbb{R}$ when Ω is a slice domain.

Slice-preserving functions are the more restrictive subclass of slice-regular functions on which classical complex methods apply slicewise. They are precisely the sections of $\mathcal{R}$ (Definition 5.1), and on them the regular product is the ordinary pointwise product, so the product is commutative (Theorem 5.2; Ghiloni–Perotti–Stoppato [7, Rem. 1.8]).

Remark 3.6 (Compactified extension). Definition 2.4 through Definition 3.5, and Theorems 3.2–3.4, are stated in the literature on uncompactified domains $\Omega \subseteq \mathbb{O}$ (and $U \subseteq \mathbb{C}$). We use them on the compactified $\mathbb{O}^* = S^8$ and on the slice Riemann spheres $\mathbb{C}_v^* = S_v^2$ (GPV’s slice Riemann manifolds [17]). The passage $\mathbb{O} \rightsquigarrow \mathbb{O}^*$, $\mathbb{C}_v \rightsquigarrow \mathbb{C}_v^*$ adjoins only the shared point $\infty = N$; it is recorded as a marked extension node (in the Lean development: cite the uncompactified statement and extend through the one-point compactification `OnePoint`).

4 The octonionic zeta function $\zeta_{\mathbb{O}^*}$

Definition 4.1 (Octonionic zeta). Here N denotes the *north pole* of S^8 : the single point at infinity ∞ adjoined in the one-point compactification $\mathbb{O}^* = \mathbb{O} \cup \{\infty\} \cong S^8$ (inverse stereographic projection; GPV [17]). It is one point with two names, so $\infty = N$; it is the only point at infinity, shared by $\mathbb{O}^*$ and by every slice Riemann sphere $\mathbb{C}_v^* = S_v^2$.

The *octonionic zeta function* $\zeta_{\mathbb{O}^*} : \mathbb{O}^* \rightarrow \mathbb{O}^*$ is defined slicewise from the compactified classical zeta (Definition 1.6) through the slice identifications $\varphi_v : \mathbb{C}^* \xrightarrow{\sim} \mathbb{C}_v^* = S_v^2$ (the $\mathbb{R}$ -algebra isomorphism $\mathbb{C} \rightarrow \mathbb{C}_v$ of Definition 2.4, extended to the Riemann spheres by $\varphi_v(\infty) = N$; GPV [17], Prop. 4.1/Thm. 4.2):

- (i) for $s \in \mathbb{O} \setminus \mathbb{R}$, with $w = \text{Im}(s)$, $v = w/|w| \in S^6$, $\gamma = |w| > 0$, $\sigma = \text{Re}(s)$, so $s = \sigma + \gamma v$:
 $\zeta_{\mathbb{O}^*}(s) := \varphi_v(\zeta_{\mathbb{C}^*}(\sigma + i\gamma))$;
- (ii) for $s \in \mathbb{R} \setminus \{1\}$: $\zeta_{\mathbb{O}^*}(s) := \zeta_{\mathbb{C}^*}(s) = \zeta(s) \in \mathbb{R}$;
- (iii) for $s = 1$ (the simple pole, Definition 1.6): $\zeta_{\mathbb{O}^*}(1) := \infty = N$;
- (iv) for $s = \infty = N$: $\zeta_{\mathbb{O}^*}(\infty) := \zeta_{\mathbb{C}^*}(\infty) = 1$.

Proposition 4.2 (Well-definedness). $\zeta_{\mathbb{O}^*} : \mathbb{O}^* \rightarrow \mathbb{O}^*$ is well-defined.

Proof. For $s \in \mathbb{O} \setminus \mathbb{R}$ the decomposition $s = \sigma + \gamma v$ with $\gamma > 0$, $v \in S^6$ is forced ($\gamma = |\text{Im } s|$, $v = \text{Im } s / |\text{Im } s|$), so the only ambiguity is the identification of the slice $\mathbb{C}_v = \mathbb{C}_{-v}$ with $\mathbb{C}^*$. That slice carries two such identifications, $\varphi_v (i \mapsto v)$ and $\varphi_{-v} (i \mapsto -v)$, under which s reads as $\sigma + i\gamma$ resp. $\sigma - i\gamma$; we must check $\varphi_v(\zeta_{\mathbb{C}^*}(\sigma + i\gamma)) = \varphi_{-v}(\zeta_{\mathbb{C}^*}(\sigma - i\gamma))$. By conjugate symmetry $\zeta_{\mathbb{C}^*}(\sigma - i\gamma) = \overline{\zeta_{\mathbb{C}^*}(\sigma + i\gamma)}$ ([15, Ch. 2]; Definition 1.6), writing $\zeta_{\mathbb{C}^*}(\sigma + i\gamma) = a + bi$ gives $\varphi_{-v}(a - bi) = a + (-b)(-v) = a + bv = \varphi_v(a + bi)$: the two sign reversals — one from conjugating the input, one from the reversed identification — cancel. The remaining cases carry no ambiguity: for real $s \neq 1$, $\zeta_{\mathbb{O}^*}(s) = \zeta(s) \in \mathbb{R} \subseteq \mathbb{C}_v^*$ for every v ; at $s = 1$ the value is the single point $\infty = N$; and at $s = \infty = N$ the value is $\zeta_{\mathbb{C}^*}(\infty) = 1 \in \mathbb{R}$, again the same for every v since each φ_v fixes $\mathbb{R}$ pointwise. $\square$

Remark 4.3 (Domain, codomain, and the point at infinity). Both domain and codomain are $\mathbb{O}^* = \mathbb{O} \cup \{\infty\} \cong S^8$, the one-point (Alexandroff) compactification of $\mathbb{O} \cong \mathbb{R}^8$ [12]. The map $\zeta_{\mathbb{O}^*}$ is slice regular on $\mathbb{O}^* \setminus \{1, \infty\}$ (Theorem 7.2); the simple pole at $s = 1$ is carried to the point at infinity $\infty = N$, which is the *image* of the pole, not a zero. We write $\mathbb{O}^*$ throughout; it is the one-point compactification, and is deliberately **not** a projective space $\mathbb{O}P^1$ (there is no quotient by scalars).

5 $\mathcal{R}$ is a commutative ring

We work on $\mathbb{O}^*$; write $\Omega_v := \mathbb{O}^* \cap \mathbb{C}_v = \mathbb{C}_v^* = S_v^2$ for the slice Riemann sphere (Definition 2.4).

Definition 5.1 (The ring of slice-preserving functions under the regular product). $\mathcal{R} := \mathcal{O}_{\mathbb{O}^*}$ is the set of all slice-regular functions $f : \mathbb{O}^* \rightarrow \mathbb{O}^*$ that are slice preserving (Definition 3.5), $f(\Omega_v) \subseteq \mathbb{C}_v^*$ for all $v \in S^6$, equipped with pointwise addition and the regular $(*)$ product.

Commutativity of $\mathcal{R}$ rests on a single slice-preserving fact from the literature: on each slice the regular product is ordinary multiplication.

Theorem 5.2 (Regular product on slice-preserving functions; [16, Rem. 2.11]). *For slice-preserving f, g (with $f(\Omega_v) \subseteq \mathbb{C}_v$), the regular product restricts on each slice to the pointwise product, $(f * g)|_{\Omega_v}(w) = f(w) \cdot g(w) \in \mathbb{C}_v$, and $f * g = g * f$, $(f * g) * h = f * (g * h)$ (via Artin, Theorem 2.2).*

Proposition 5.3. $(\mathcal{R}, +, *)$ is a commutative ring with identity $1 \in \mathcal{R}$.

Proof. Closure under $+$ is pointwise. For $*$: fixing a slice $\mathbb{C}_K$, the product reduces to pointwise multiplication on $\mathbb{C}_K$ by Theorem 5.2, which is commutative because $\mathbb{C}_K$ is a commutative associative subalgebra; associativity and distributivity are inherited slicewise, and a slice-regular function is determined by its restriction to any one slice (Theorem 3.3). The constant 1 is the identity. $\square$

Proposition 5.4 ($\mathcal{R}$ is an integral domain). *If $f, g \in \mathcal{R}$ and $f * g = 0$, then $f = 0$ or $g = 0$.*

Proof. $(f * g)|_{\Omega_K}(w) = f(w) \cdot g(w)$ in $\mathbb{C}_K \cong \mathbb{C}$ (Theorem 5.2). The restrictions $f|_{\Omega_K}, g|_{\Omega_K}$ are holomorphic on the connected slice; by the classical identity principle the pointwise product vanishes identically iff one factor does. By Theorem 3.3 the corresponding global function vanishes. $\square$

6 The slice-preserving analytic toolkit on the compactified octonions

We collect the slice-preserving analytic facts used in Part 3 as cited statements: the exponential, its logarithm manifold, its degenerate fibre, the slice/stem square and I -independent lift, and the winding lift. Each is established in the literature on the uncompactified $\mathbb{O}$ and used here on $\mathbb{O}^* = S^8$ and the slice Riemann spheres $\mathbb{C}_v^* = S_v^2$ (the passage is recorded in Remark 3.6; the compactified slice-manifold structure itself is GPV's [17]).

Theorem 6.1 (The slice exponential; [21, §3]; [17, Prop. 5.1]; [1, Rem. 2.23]). *The exponential $\exp : \mathbb{O} \rightarrow \mathbb{O} \setminus \{0\}$, $\exp(q) = \sum_{k \geq 0} q^k / k!$ (powers well defined by Corollary 2.3), is a slice-regular, slice-preserving entire function (Definition 3.5), given on each slice by*

$$\exp(x + Iy) = e^x(\cos y + I \sin y), \quad x, y \in \mathbb{R}, \quad I \in S^6.$$

Consequently $|\exp q| = e^{\operatorname{Re} q}$ depends only on the real part, and $\exp$ is nowhere zero.

Theorem 6.2 (The logarithm manifold $E_{\mathbb{O}}^+$; [17, Prop. 5.1, Rem. 5.2, Def. 5.3, Prop. 5.4, Def. 5.5]; [21, §3]). *Let $T : \mathbb{O} \rightarrow \mathbb{O} \times \operatorname{Im} \mathbb{O}$, $T(x + Iy) = (\sinh x \cos y + I \sinh x \sin y, Iy)$, and $\mathbb{O}^+ = \{q : \operatorname{Re} q > 0\}$. The logarithm manifold is $E_{\mathbb{O}}^+ := T(\mathbb{O}^+)$, the semi-helicoidal hypercomplex 8-manifold; equivalently it is the image of the exponential graph, parametrised by the $E_{\mathbb{O}}^+$ -exponential*

$$E : \mathbb{O} \longrightarrow E_{\mathbb{O}}^+ \subset \mathbb{O} \times \operatorname{Im} \mathbb{O}, \quad E(x + Iy) = (\exp(x + Iy), Iy) = (e^x \cos y + Ie^x \sin y, Iy),$$

an immersion and a diffeomorphism, with inverse the $E_{\mathbb{O}}^+$ -logarithm $L : E_{\mathbb{O}}^+ \rightarrow \mathbb{O}$, $L(q, p) = \log |q| + p$ (here $p = \text{Arg}(q)$). Writing π for the projection to the first factor, $\pi \circ E = \exp$ (Theorem 6.1); but $\pi : E_{\mathbb{O}}^+ \rightarrow \mathbb{O} \setminus \{0\}$ is not a covering and not an open map, precisely because $\exp$ is not open. Thus $E_{\mathbb{O}}^+$ is an adapted blow-up of $\mathbb{O}$ along the real axis: it unwraps the multivalued logarithm into the single-valued L , and a branch of the hypercomplex logarithm is $\log_{\mathbb{O}} = L \circ (\pi|_{\Omega})^{-1}$ on any path-connected $\Omega \subset E_{\mathbb{O}}^+$ on which $\pi|_{\Omega}$ is injective.

Lemma 6.3 (The degenerate set of the exponential; its fibre over a real value). *Sourced.* With π the first-factor projection, $\pi \circ E = \exp$ ([17, Rem. 5.2(a)] — the commuting triangle); and, unlike the complex setting, $\pi : E_{\mathbb{O}}^+ \rightarrow \mathbb{O}$ is not a covering and not an open map, because $\exp$ is not open: it has a non-empty degenerate set consisting of spheres ([17, Rem. 5.2(b)], verbatim). The direction $I(q)$ admits no continuous extension to any point of $\mathbb{R}$ ([21, Rem. 2.1]).

Proved from the slice form. For $r > 0$,

$$\exp^{-1}(-r) = \{ \log r + I(2k+1)\pi : I \in S^6, k \in \mathbb{Z} \} :$$

by Theorem 6.1, $\exp(x + Iy) = e^x \cos y + Ie^x \sin y = -r$ forces $e^x \sin y = 0$, hence $y \in \pi\mathbb{Z}$; negativity of the value forces $y = (2k+1)\pi$ and $x = \log r$; the direction I is unconstrained. The fibre is thus indexed by the single real level $\log r = \log |-r|$ and, within it, by the odd winding indices $2k+1$ carried as band data (the winding lift, Theorem 6.6; [21, Cor. 5.21]). The fibre formula itself appears, stated without proof as Preface motivation, in [17] (p. 972); the slice-form derivation above is the load-bearing statement.

Remark 6.4 (“Concentric” is translation vocabulary). Geometrically the fibre of Lemma 6.3 is a family of 6-spheres of radii $\pi, 3\pi, 5\pi, \dots$ sharing the centre $\log r$ — concentric. The construction of Part 3 never uses this metric fact: the index relations enter only through the commuting triangle $\pi \circ E = \exp$, the level, and the winding. The word belongs to the downstream translation (Corollary 10.9), where one component of the static base is read as one real centre.

Definition 6.5 (Slice restriction and the I -independent lift; [16, Rem. 2.11]; [4, §3.3, §3.7]; [7, Def. 2.1, Prop. 2.2]). A slice-preserving section A carries each slice Riemann sphere into itself,

$$A(\mathbb{C}_v^*) \subseteq \mathbb{C}_v^* = S_v^2 \quad (v \in S^6),$$

so its restriction to a slice is a holomorphic self-map of that slice sphere,

$$A_v := \varphi_v^{-1} \circ A \circ \varphi_v : \mathbb{C}^* \longrightarrow \mathbb{C}^*,$$

through the identification $\varphi_v : \mathbb{C}^* \xrightarrow{\sim} \mathbb{C}_v^*$ of Definition 2.4. The I -independent lift is the fact that this holomorphic stem is the *same* for every v : there is one holomorphic $A_{\circ} : \mathbb{C}^* \rightarrow \mathbb{C}^*$ with

$$A|_{\mathbb{C}_v^*} = \varphi_v \circ A_{\circ} \circ \varphi_v^{-1} \quad \text{for all } v \in S^6$$

(Wang [16, Rem. 2.11]; Bisi–Winkelmann [4, §3.7]). Equivalently, A is *intrinsic*, $A(q^c) = A(q)^c$, and is induced ($A = \mathcal{I}(F)$) by a single stem function $F = F_1 + \iota F_2 : D \rightarrow \mathbb{O}_{\mathbb{C}}$ on the symmetric $D \subset \mathbb{C}$ whose components F_1, F_2 are $\mathbb{R}$ -valued (the stem-function formalism, Ghiloni–Perotti [7, Def. 2.1, Prop. 2.2]):

$$A(x + Iy) = F_1(x + iy) + I F_2(x + iy), \quad F_1, F_2 \text{ } \mathbb{R}\text{-valued.}$$

It is this single I -independent slice stem that makes the section functor Φ G_2 -equivariant, hence blind to the G_2 -orbit direction (Definition 9.4, Theorem 9.7).

Theorem 6.6 (The winding lift; [21, Def. 3.4, Def. 4.1, Prop. 4.2, Def. 5.11]). *Let $\gamma : [a, b] \rightarrow \mathbb{O} \setminus \{0\}$ be a path (on each slice $\mathbb{C}_v$ this is the classical complex continuation). A continuation of the hypercomplex logarithm along γ is a path $\tilde{\gamma} : [a, b] \rightarrow \mathbb{O}$ with $\exp \circ \tilde{\gamma} = \gamma$, and a lift of γ to the logarithm manifold $E_{\mathbb{O}}^+$ (Theorem 6.2) is a path $\Gamma : [a, b] \rightarrow E_{\mathbb{O}}^+$ with $\text{pr}_1 \circ \Gamma = \gamma$:*

$$\begin{array}{ccc} & \nearrow \tilde{\gamma} & \mathbb{O} \\ [a, b] & \xrightarrow{\gamma} & \mathbb{O} \setminus \{0\} \\ & \uparrow \exp & \end{array} \qquad \begin{array}{ccc} & \nearrow \Gamma & E_{\mathbb{O}}^+ \\ [a, b] & \xrightarrow{\gamma} & \mathbb{O} \setminus \{0\} \\ & \downarrow \text{pr}_1 & \end{array}$$

The two data are equivalent: a lift Γ exists if and only if a continuation $\tilde{\gamma}$ exists, and they correspond by

$$\tilde{\gamma} = L \circ \Gamma, \quad \Gamma = (\exp \circ \tilde{\gamma}, \text{Im } \tilde{\gamma}),$$

where L is the $E_{\mathbb{O}}^+$ -logarithm of Theorem 6.2 ([21, Prop. 4.2]). For a loop $\gamma : S^1 \rightarrow \mathbb{O} \setminus \{0\}$ the lift is taken on the universal cover $\exp : i\mathbb{R} \rightarrow S^1$, as a continuous $\Gamma : i\mathbb{R} \rightarrow E_{\mathbb{O}}^+$ with $\text{pr}_1 \circ \Gamma = \gamma \circ \exp$ ([21, Def. 5.11]):

$$\begin{array}{ccc} i\mathbb{R} & \xrightarrow{\Gamma} & E_{\mathbb{O}}^+ \\ \exp \downarrow & & \downarrow \text{pr}_1 \\ S^1 & \xrightarrow{\gamma} & \mathbb{O} \setminus \{0\} \end{array}$$

Proposition 6.7 (Tameness, existence, and the winding number; [21, Def. 4.20, Cor. 5.21, Cor. 5.22]).

The lift of Theorem 6.6 is governed by the companion structure of γ .

- (i) *Uniqueness (tameness).* γ is tame when it admits a unique companion imaginary-unit field; its lift is then unique ([21, Def. 4.20]).
- (ii) *Existence and closure.* A lift of a loop γ exists if and only if the signature satisfies $\sigma(\gamma|_{[\xi_l, \xi_{l+1}]}) \in \{0, -1\}$ on each obstruction interval; and if it exists, the lift is itself a loop ([21, Cor. 5.13]; pinned against the arXiv text — the earlier citation to Cor. 5.22 is superseded).
- (iii) *Winding number.* For an untwisted tame loop with even circular signature σ^c , the winding number is $\omega = |\sigma^c|/2$ ([21, Cor. 5.21]).

7 Slice preservation and symmetries

Theorem 7.1 (Slice preservation). $\zeta_{\mathbb{O}^*}$ is slice preserving (Definition 3.5): $\zeta_{\mathbb{O}^*}(\mathbb{C}_v^*) \subseteq \mathbb{C}_v^*$ for every $v \in S^6$.

Proof. Fix $v \in S^6$ and $s \in \mathbb{C}_v^*$. If $s \in \mathbb{C}_v \setminus \mathbb{R}$ then $s = \sigma \pm \gamma v$ and $\zeta_{\mathbb{O}^*}(s) = \varphi_v(\zeta_{\mathbb{C}^*}(\sigma \pm i\gamma)) \in \varphi_v(\mathbb{C}^*) = \mathbb{C}_v^*$ (Definition 4.1(i)). If $s \in \mathbb{R} \setminus \{1\}$ then $\zeta_{\mathbb{O}^*}(s) = \zeta(s) \in \mathbb{R} \subseteq \mathbb{C}_v^*$; while $\zeta_{\mathbb{O}^*}(1) = \infty \in \mathbb{C}_v^*$ and $\zeta_{\mathbb{O}^*}(\infty) = 1 \in \mathbb{C}_v^*$. In every case the value remains in $\mathbb{C}_v^*$. $\square$

Theorem 7.2 ($\zeta_{\mathbb{O}^*}$ is a section of $\mathcal{R}$). $\zeta_{\mathbb{O}^*} \in \mathcal{R}$.

Proof. On each slice the restriction is the compactified classical zeta read through the isomorphism φ_v , namely $\zeta_{\mathbb{O}^*}|_{\mathbb{C}_v^*} = \varphi_v \circ \zeta_{\mathbb{C}^*} \circ \varphi_v^{-1}$, holomorphic into $\mathbb{C}_v^*$ away from the single pole. Equivalently $\zeta_{\mathbb{O}^*} = \text{ext}(\zeta)$ is the slice-regular extension of the holomorphic, conjugation-symmetric ζ on $\mathbb{C} \setminus \{1\}$ (Theorem 3.4, [5, Thm. 5.1.5]); hence $\zeta_{\mathbb{O}^*}$ is slice regular on $\mathbb{O}^* \setminus \{1, \infty\}$ (slice regularity is slicewise

holomorphy, Definition 3.1). By Theorem 7.1 it is slice preserving, $\zeta_{\mathbb{O}^*}(\mathbb{C}_v^*) \subseteq \mathbb{C}_v^*$. A slice-regular, slice-preserving function $\mathbb{O}^* \rightarrow \mathbb{O}^*$ is by definition a section of $\mathcal{R} = \mathcal{O}_{\mathbb{O}^*}$ (Definition 5.1); the regular product on such sections restricts on each slice to the ordinary product (Theorem 5.2, Wang [16, Rem. 2.11]). Finally $\zeta_{\mathbb{O}^*}$ is not a unit, since it has zeros (Theorem 8.2). $\square$

Definition 7.3 (The automorphism group G_2). $G_2 = \text{Aut}(\mathbb{O})$ is the 14-dimensional compact, connected, simply connected Lie group of $\mathbb{R}$ -algebra automorphisms of $\mathbb{O}$. Each $g \in G_2$ fixes 1, hence the real axis pointwise, and acts on $\text{Im } \mathbb{O} \cong \mathbb{R}^7$ as an element of $SO(7)$; thus $g \cdot (\sigma + \gamma v) = \sigma + \gamma g(v)$.

Theorem 7.4 (G_2 -equivariance). $\zeta_{\mathbb{O}^*}(g \cdot s) = g \cdot \zeta_{\mathbb{O}^*}(s)$ for all $g \in G_2$, $s \in \mathbb{O}$.

Proof. For $s \in \mathbb{R}$, $g \cdot s = s$ and $\zeta_{\mathbb{O}^*}(s) \in \mathbb{R}$ is fixed, so both sides equal $\zeta_{\mathbb{O}^*}(s)$. For a non-real $s = \sigma + \gamma v$ ($v \in S^6$, $\gamma > 0$), g carries the slice $\mathbb{C}_v$ to the slice $\mathbb{C}_{g(v)}$ by the isometric relabelling $\varphi_{g(v)} = g \circ \varphi_v$ (Definition 7.3). Since $\zeta_{\mathbb{O}^*}$ is defined slicewise by the same $\zeta_{\mathbb{C}^*}$ read through φ_v (Definition 4.1(i)),

$$\zeta_{\mathbb{O}^*}(g \cdot s) = \varphi_{g(v)}(\zeta_{\mathbb{C}^*}(\sigma + i\gamma)) = g(\varphi_v(\zeta_{\mathbb{C}^*}(\sigma + i\gamma))) = g \cdot \zeta_{\mathbb{O}^*}(s).$$

$\square$

Theorem 7.5 ([3]). G_2 acts transitively on S^6 with stabilizer $SU(3)$: $SU(3) \hookrightarrow G_2 \twoheadrightarrow S^6$.

Remark 7.6 (The G_2 -action extends to $\mathbb{O}^* = S^8$). The action above is stated on $\mathbb{O}$, but $\mathcal{H}_1$ (Part 3) runs on the compactified $\mathbb{O}^* = S^8$. Each $g \in G_2$ fixes the real axis pointwise (an automorphism fixes 1, hence all of $\mathbb{R}$) and acts on $\text{Im } \mathbb{O} = \mathbb{R}^7$ as an element of $SO(7)$; in particular g is a linear isometry of $\mathbb{R}^8 = \mathbb{O}$, hence proper, and therefore extends uniquely to a homeomorphism of the one-point compactification $\mathbb{O}^* = \mathbb{O} \cup \{\infty\} = S^8$ fixing the point at infinity $\infty = N$ (the one-point compactification is functorial on proper maps; in Lean, `OnePoint`). Thus $G_2 \curvearrowright \mathbb{O}^*$ fixes the compactified real axis $\mathbb{R} \cup \{\infty\}$ — a great circle through N — pointwise, and acts on each direction-sphere S^6 exactly as in Theorem 7.5. This is the action on which $\mathcal{H}_1$ is built, with $\mathcal{R}$ its ring of invariants over $\mathbb{O}^*$.

8 Zero geometry and the equivalence

Theorem 8.1 (Zero Equivalence Theorem). Let $\rho = \sigma + i\gamma \in \mathbb{C}^*$. For every $v \in S^6$,

$$\zeta_{\mathbb{O}^*}(\sigma + \gamma v) = 0 \iff \zeta_{\mathbb{C}^*}(\rho) = 0.$$

Equivalently: a non-real $s \in \mathbb{O}^*$ is a zero of $\zeta_{\mathbb{O}^*}$ if and only if its slice coordinate $\varphi_v^{-1}(s)$ is a zero of $\zeta_{\mathbb{C}^*}$; by Lemma 1.7 these are exactly the nontrivial zeros of the classical ζ .

Proof. By Definition 4.1(i), $\zeta_{\mathbb{O}^*}(\sigma + \gamma v) = \varphi_v(\zeta_{\mathbb{C}^*}(\sigma + i\gamma))$. The slice identification $\varphi_v : \mathbb{C}^* \rightarrow \mathbb{C}_v^* = S_v^2$ is an isomorphism of slice Riemann spheres (the $\mathbb{R}$ -algebra isomorphism $\mathbb{C} \rightarrow \mathbb{C}_v$ of Definition 2.4, extended by $\varphi_v(\infty) = N$; GPV [17]). An isomorphism is injective and sends 0 to 0, so $\varphi_v(z) = 0 \iff z = 0$; with $z = \zeta_{\mathbb{C}^*}(\rho)$ this gives both implications. (Slice preservation, Theorem 7.1, is what places the value in $\mathbb{C}_v^*$ so that φ_v^{-1} applies on the non-real side.) The final clause is Lemma 1.7. $\square$

Theorem 8.2 (Nontrivial zeros are infinitely many disjoint 6-spheres). For a nontrivial zero $\rho = \sigma + i\gamma$ of $\zeta_{\mathbb{C}^*}$ (so $\gamma > 0$) set

$$S_\rho = \{\sigma + \gamma v : v \in S^6\} = \sigma + \gamma S^6 \subset \mathbb{O}.$$

Then:

- (i) S_ρ is a 6-sphere — the round sphere of radius γ centred at the real point σ — equivalently the G_2 -orbit of any one of its points $\sigma + \gamma v_0$ (Theorem 7.4; G_2 acts transitively on S^6 with stabiliser $\mathrm{SU}(3)$, Theorem 7.5, Baez [3]).
- (ii) every point of S_ρ is a zero of $\zeta_{\mathbb{O}^*}$, and conjugate zeros give the same sphere, $S_\rho = S_{\bar{\rho}}$;
- (iii) the nontrivial-zero set is the disjoint union $Z_{\mathbb{O}^*}^{\mathrm{nt}} = \bigsqcup_{[\rho]} S_\rho$ over conjugate pairs $[\rho] = \{\rho, \bar{\rho}\}$, with distinct pairs giving disjoint spheres;
- (iv) there are infinitely many such spheres (Corollary 1.4; also Hardy [11], Theorem 1.5).

Proof. (i) The map $v \mapsto \sigma + \gamma v$ is the affine map $x \mapsto \sigma + \gamma x$ of $\mathrm{Im} \mathbb{O} \cong \mathbb{R}^7$ restricted to S^6 ; with $\gamma > 0$ its image is the round 6-sphere of radius γ centred at σ . By G_2 -equivariance (Theorem 7.4) and transitivity of G_2 on S^6 (Theorem 7.5), $G_2 \cdot (\sigma + \gamma v_0) = \{\sigma + \gamma g(v_0) : g \in G_2\} = S_\rho$. So S_ρ is S^6 — scaled by γ , translated to σ — realized as a single G_2 -orbit; no embedding or diffeomorphism is invoked.

(ii) For any $v \in S^6$, $\zeta_{\mathbb{O}^*}(\sigma + \gamma v) = \varphi_v(\zeta_{\mathbb{C}^*}(\rho)) = \varphi_v(0) = 0$ by the Zero Equivalence Theorem 8.1. And $S_{\bar{\rho}} = \{\sigma + \gamma(-v) : v \in S^6\} = S_\rho$, since $v \mapsto -v$ is a bijection of S^6 .

(iii) For “ $\subseteq$ ”: a non-real zero $s = \sigma + \gamma v$ of $\zeta_{\mathbb{O}^*}$ forces $\zeta_{\mathbb{C}^*}(\sigma + i\gamma) = 0$ by Theorem 8.1, so $s \in S_\rho$ with $\rho = \sigma + i\gamma$ nontrivial (Lemma 1.7); “ $\supseteq$ ” is (ii). Disjointness: if $\sigma + \gamma v = \sigma' + \gamma' w$, uniqueness of the real/imaginary parts gives $\sigma = \sigma'$ and $\gamma v = \gamma' w$, whence $\gamma = \gamma'$ (taking norms, $|v| = |w| = 1$) and $\{\rho', \bar{\rho}'\} = \{\rho, \bar{\rho}\}$; so two spheres coincide or are disjoint.

(iv) There are infinitely many spheres because ζ has infinitely many nontrivial zeros (Corollary 1.4; also Hardy, Theorem 1.5). $\square$

Theorem 8.3 (The equivalence: concentricity $\Leftrightarrow$ RH). *The following are equivalent:*

- (a) the Riemann Hypothesis;
- (b) the nontrivial-zero 6-spheres $\{S_\rho\}$ of $\zeta_{\mathbb{O}^*}$ (Theorem 8.2) are concentric — they share a single common centre, necessarily a point of the real axis.

When they hold, the common centre is $\frac{1}{2}$.

Proof. Each S_ρ is centred at the real point σ_ρ , where $\rho = \sigma_\rho + i\gamma_\rho$ (Theorem 8.2(i)).

(a) $\Rightarrow$ (b): under RH every $\sigma_\rho = \frac{1}{2}$, so all spheres share the centre $\frac{1}{2}$.

(b) $\Rightarrow$ (a): suppose the spheres share a common centre $c \in \mathbb{R}$. By the functional equation (Theorem 1.1), if ρ is a nontrivial zero then so is $1 - \bar{\rho} = (1 - \sigma_\rho) + i\gamma_\rho$, whose sphere $S_{1-\bar{\rho}}$ is centred at $1 - \sigma_\rho$. Concentricity forces $\sigma_\rho = c = 1 - \sigma_\rho$, hence $\sigma_\rho = \frac{1}{2}$; as ρ was arbitrary this is RH, and the common centre is $\frac{1}{2}$. $\square$

Remark 8.4 ($\frac{1}{2}$ is downstream of concentricity). The value $\frac{1}{2}$ enters *only* in (b) $\Rightarrow$ (a), and only through the functional equation pinning the shared hyperplane once concentricity is known. Part 3 establishes the one-component collapse — that the residue- $\mathbb{C}$ zero spheres of A all lie in one connected component of the Grothendieck construction $\mathcal{T}_A$ (Definition 9.9), concentricity then following by the translation of Corollary 10.9 — with no reference to $\frac{1}{2}$, to coordinates, or to RH; the value is then supplied here, downstream, by the classical functional equation alone.

Part III

The Categorical Construction and the Concentricity Theorem

We build the translation groupoid $\mathcal{H}_1 = G_2 \ltimes \mathbb{O}^*$ (whose orbits of non-real points are the residue- $\mathbb{C}$ zero 6-spheres), the slice world $\mathcal{S}_2$ (the codomain $\mathbb{O}^*$ by its slice decomposition, carrying the band $U(1)$ and the direction G_2), the section functor $\Phi : \mathcal{H}_1 \rightarrow \mathcal{S}_2$, the base $\mathcal{B}$ of the exponential's degenerate set, and the Grothendieck construction $\mathcal{T}_A = \int_{\mathcal{B}} F$ of the band $F = U(1)$ over $\mathcal{B}$. Hypotheses C1–C4 assemble the transport of $\mathcal{T}_A$; the residue- $\mathbb{C}$ zero 6-spheres of A arrive as the degenerate fibre of that transport — an output, never an input — and the Concentricity Theorem is that they lie in a single connected component of $\mathcal{T}_A$, computed at π_0 . The categorical objects are anchored to the literature (Quillen [24], Thomason [23], Riehl [25], Goerss–Jardine [20], Vistoli [26], Stacks [18]) and to Mathlib terms; the slice-preserving analytic facts of Part 2 enter as cited inputs.

9 Two worlds and the section between them

The compactified octonions and the point at infinity

Definition 9.1 (The compactified octonions $\mathbb{O}^* = S^8$). Let $\mathbb{O}^* = \mathbb{O} \cup \{\infty\} = S^8$ be the Alexandroff one-point compactification of $\mathbb{O} = \mathbb{R}^8$. Concretely (GPV [17], Prop. 4.1 and Thm. 4.2, for $m = \dim_{\mathbb{R}} K \in \{4, 8\}$): the inverse stereographic projections from the north and south poles,

$$f(x + Iy) = \left(\frac{2(x+Iy)}{1+x^2+y^2}, \frac{-1+x^2+y^2}{1+x^2+y^2} \right), \quad h(x + Iy) = \left(\frac{2(x-Iy)}{1+x^2+y^2}, \frac{1-x^2-y^2}{1+x^2+y^2} \right),$$

give a conformal atlas $\{(K, f), (K, h)\}$ endowing S^m with the structure of a slice (quaternionic/octonionic) Riemann manifold, with transition map $h^{-1} \circ f(q) = \bar{q}/q^2 = 1/q$, itself slice regular and slice preserving. The adjoined point is the north pole N ; this is the point at infinity. In Lean the underlying compactification is `Mathlib.Topology.Compactification.OnePoint`, $\mathbb{O}^* = \text{OnePoint } \mathbb{O}$ with $\infty = \text{OnePoint.infty}$, and the homeomorphism $\mathbb{O}^* \cong S^8$ is `onePointEquivSphereOfFinrankEq` applied to $\dim_{\mathbb{R}} \mathbb{O} = 8$ (`Mathlib.Topology.Compactification.OnePoint.Sphere`). The compactification is what makes the value of a section at its real pole a well-defined point of $\mathbb{O}^*$ — the point at infinity $\infty = N$. There is no base poset: the two worlds are built directly on $\mathbb{O}^*$ as translation groupoids (Definition 9.6).

Remark 9.2 (Compactifying the cited results). Most of the slice-preserving / slice-regular results cited below (GPV, AdF, Wang, Bisi–Winkelmann) are stated on uncompactified $\mathbb{O} = \mathbb{R}^8$ or domains therein; the construction lives on $\mathbb{O}^* = S^8$. Each citation is used in its compactified form, obtained either directly (GPV's Riemann-manifold statements [17] are already on S^m) or by transporting the uncompactified statement across the stereographic atlas of Definition 9.1; in the Lean development this transport is a marked derivation node over `OnePoint`, not an assumption.

Remark 9.3 (The two north pole objects). There are two north pole objects, one in each world. In $\mathcal{H}_1$ the north pole N of $\mathbb{O}^*$ is a G_2 -fixed object: its own connected component, with $\text{Aut}(N) = G_2$, carrying no universal property. In the slice world $\mathcal{S}_2$ the north pole is the object

$$\mathbf{n} := \infty = N \text{ regarded as an object of } \mathcal{S}_2,$$

the common point at infinity of the slice Riemann spheres, fixed by band and direction alike. They are objects of *different* groupoids, related by the section through C1: the pole of A has value ∞ , so the section functor carries the pole point to $\mathfrak{n}$ (Remark 9.8). The residue field of a closed zero sorts the picture: residue- $\mathbb{R}$ (the G_2 -fixed locus $\mathbb{R} \cup \{N\}$, classically trivial) versus residue- $\mathbb{C}$ (the 6-sphere orbits, classically nontrivial; the dictionary is Part 2 and Theorem 8.2).

What a slice-preserving section does

Definition 9.4 (The section's slice data and equivariance). Let $A \in \mathcal{R}$. By the I -independent lift (Definition 6.5; Wang [16, Rem. 2.11], Bisi–Winkelmann [4, §3.7]) there is one holomorphic stem with $\mathbb{R}$ -valued components,

$$A(x + Iy) = F_1(x + iy) + I F_2(x + iy), \quad F_1, F_2 \text{ } \mathbb{R}\text{-valued,}$$

the same for every $I \in S^6$. Three consequences used below:

- (i) *Slice preservation of values.* $A(\mathbb{C}_I^*) \subseteq \mathbb{C}_I^*$ for every I : the value at a point lies on that point's own slice sphere (Definition 5.1).
- (ii) *G_2 -equivariance.* For $g \in G_2$ and non-real $q = x + Iy$,

$$A(g \cdot q) = F_1 + g(I) F_2 = g \cdot (F_1 + I F_2) = g \cdot A(q),$$

since $g \cdot (x + Iy) = x + g(I)y$ (Definition 7.3) and F_1, F_2 are real; for $q \in \mathbb{R} \cup \{N\}$ both sides are G_2 -fixed, because $A(\mathbb{R}) \subseteq \mathbb{R}$ and $A(N) \in \bigcap_I \mathbb{C}_I^* = \mathbb{R} \cup \{N\}$ by (i). (For $\zeta_{\mathbb{O}^*}$ this is Theorem 7.4; the computation above is the same one, run for an arbitrary intrinsic section.)

- (iii) *Blindness to the sphere direction.* On a G_2 -orbit $S_{(\sigma, \gamma)}$ the stem coordinates $(F_1, F_2)(\sigma + i\gamma)$ are constant; in particular if A vanishes at one point of the orbit it vanishes on the whole orbit (Lemma 10.2).

No sphere-fibration and no tower enter: a slice-preserving section carries one holomorphic stem per slice, and the direction S^6 is traversed only by the G_2 -morphisms already present in both worlds.

Remark 9.5 (Role of C4). Condition C4 — infinitely many residue- $\mathbb{C}$ zeros — matches the class to its infinite structure (the infinite Euler family of C2, the infinite Weierstrass divisor of C3) and gives the collapse its force: the degenerate fibre of the transport is an infinite family, so that whether it lies in one connected component of $\mathcal{T}_A$ (Definition 9.9) is an infinite alignment, decided by Theorem 10.5. There is one connected component or there is not; nothing intermediate.

The two worlds as translation groupoids

Definition 9.6 (The two translation groupoids). Recall the *translation groupoid* (action groupoid) of a group action $G \curvearrowright X$: the category $G \ltimes X$ whose objects are the points of X and whose morphisms $x \rightarrow y$ are the elements $g \in G$ with $g \cdot x = y$; it is a groupoid, and $\pi_0(G \ltimes X) = X/G$ is the orbit set (Quillen [24, §1]: components of a category $\leftrightarrow$ components of its classifying space; in Lean, `CategoryTheory.ActionCategory`).

The *octonionic world* is the translation groupoid $\mathcal{H}_1 = G_2 \ltimes \mathbb{O}^*$ of $G_2 = \text{Aut}(\mathbb{O})$ acting on $\mathbb{O}^* = S^8$ (Definitions 9.1 and 7.3; Remark 7.6). Its connected components are the G_2 -orbits, of two kinds (Baez [3]; Theorem 7.5; Remark 7.6):

- the fixed points, forming the compactified real axis $\mathbb{R} \cup \{N\}$, of residue field $\mathbb{R}$; and
- for each real σ and each radius $\gamma > 0$, the direction 6-sphere $S_{(\sigma, \gamma)} = \{\sigma + \gamma v : v \in S^6\}$, of residue field $\mathbb{C}$ — the G_2 -orbit of any of its points, labelled by its real centre σ and radius γ .

The *slice world* $\mathcal{S}_2$ organizes the *codomain* $\mathbb{O}^*$ of a section by its slice decomposition. Every slice contains the real axis by the definition of a slice, $\mathbb{C}_I = \text{Span}_{\mathbb{R}}(1, I)$, two slices meeting exactly in $\mathbb{R}$ (Definition 2.4; [7, §1–§2]), and the compactification adjoins the single shared point $\infty = N$ (Definition 9.1; GPV [17]), so

$$\mathbb{O}^* = \bigcup_{I \in S^6} \mathbb{C}_I^*, \quad \mathbb{C}_I^* \cap \mathbb{C}_J^* = \mathbb{R} \cup \{N\} \quad (J \neq \pm I).$$

$\mathcal{S}_2$ is the groupoid whose objects are the points of $\mathbb{O}^*$ so decomposed, and whose morphisms are generated by

- the *direction* isomorphisms: $g : w \rightarrow g \cdot w$ for $g \in G_2$, carrying $\mathbb{C}_I^*$ to $\mathbb{C}_{g(I)}^*$ and fixing $\mathbb{R} \cup \{N\}$ pointwise (Remark 7.6); and
- the *band* automorphism families: on each slice sphere $\mathbb{C}_I^*$, the phases $u_\theta : z \mapsto e^{I\theta} z$ — the Möbius self-maps of S_I^2 fixing the value-origin 0, the point at infinity N , and the modulus.

The objects 0 and N are fixed by every generating morphism. No identification of distinct objects is made here: the two worlds share the underlying points of $\mathbb{O}^*$ and differ in their acting structure (G_2 alone, versus direction together with the band). This is exactly the structure the base $\mathcal{B}$ already carries (Definition 9.9); nothing is added there.

The section functor

Theorem 9.7 (The section functor Φ). *For a section $A \in \mathcal{R}$, the assignment*

$$\Phi(q) = A(q) \text{ on objects,} \quad \Phi(g : q \rightarrow g \cdot q) = (g : A(q) \rightarrow A(g \cdot q)) \text{ on morphisms,}$$

is a functor of groupoids $\Phi : \mathcal{H}_1 \rightarrow \mathcal{S}_2$ — the section functor.

Proof. This is the standard fact that an equivariant map of G -sets induces a functor of the associated translation groupoids, applied to A . On objects, $A(q)$ is a point of $\mathbb{O}^*$ lying on the slice sphere of q (Definition 9.4(i)), hence an object of $\mathcal{S}_2$. On morphisms, G_2 -equivariance $A(g \cdot q) = g \cdot A(q)$ (Definition 9.4(ii)) says precisely that g is a direction morphism $A(q) \rightarrow A(g \cdot q)$ of $\mathcal{S}_2$ (Definition 9.6), so $\Phi(g)$ is well-typed. Functoriality is immediate: $\Phi(e) = \text{id}_{A(q)}$, and for composable g, h the composite of the direction morphisms $g : A(q) \rightarrow A(gq)$ and $h : A(gq) \rightarrow A(hgq)$ is $hg : A(q) \rightarrow A(hgq) = \Phi(hg)$. On each slice the underlying value assignment is the holomorphic self-map A_v of S_v^2 , compatible with the ring structure because the regular product restricts to ordinary multiplication on the commutative slice (Theorem 5.2, Proposition 5.3). $\square$

Remark 9.8 (What Φ collapses, and the cone from the pole). The collapse belongs to this functor, not to the codomain. By Definition 9.4(iii), Φ carries each residue- $\mathbb{C}$ zero-sphere of A — a 6-sphere component of $\mathcal{H}_1$ (Lemma 10.2) — onto the *single* object $0 \in \mathcal{S}_2$, the value-origin, fixed by band and direction; and by C1 it carries the pole point onto the single object $\mathfrak{n}$ (Remark 9.3). Thus the pole pulls a cone in the value world: every path of $\mathcal{H}_1$ into the pole maps to a value path into the slice-world north pole. The differing centres and radii of the zero-spheres are not remembered by Φ 's object map; they are remembered by the *transport* over the base (Definition 9.9), which is where the theorem reads them.

The base of the degenerate set and the Grothendieck construction

Definition 9.9 (The base $\mathcal{B}$ of the degenerate set, the band F , and $\mathcal{T}_A = \int_{\mathcal{B}} F$). *The base.* The exponential's degenerate fibre over a real value is the level-indexed family of Lemma 6.3: $\exp^{-1}(-r) = \{\log r + I(2k+1)\pi\}$, one real level $\log r$, winding as band data — a structure of the space itself, absent over $\mathbb{C}$ ([17, Rem. 5.2(b)]). The *base* $\mathcal{B}$ is the groupoid of these levels: one object for each real level c , with the direction automorphisms G_2 at each level. GPV's $\log |\cdot|$ is a label on objects, never an operation. Distinct levels are distinct objects with *no* morphisms between them, so the level is constant along every zigzag and $\pi_0(\mathcal{B})$ is the set of levels by construction — connectivity here is the groupoid's (reachability by declared morphisms, Definition 9.6), not the ambient topology of $\mathbb{O}$.

The band. Let $F : \mathcal{B} \rightarrow \text{Grpd}$ send each level to the band groupoid $U(1)$ (one object; automorphism group the phases) and each direction morphism to the identity of the band. The winding index $2k+1$ of the fibre is carried by the band as an automorphism datum within a component — the loops of $F(c)$ realize the winding number $\omega = |\sigma^c|/2$ of the lifted value-loops ([21, Cor. 5.21]; Proposition 6.7) — and is never an object label of $\mathcal{B}$.

The total object. The *Grothendieck construction* of F is the category

$$\mathcal{T}_A = \int_{\mathcal{B}} F,$$

with objects the pairs (c, x) , x in the band over c , and morphisms the pairs (f, φ) moving the base and transporting the fibre (Thomason [23]; Vistoli [26, §3.1]; Stacks [18, Tag 0049]; in Lean, `CategoryTheory.Grothendieck`). By Thomason's theorem its classifying space models the homotopy colimit, $B(\int_{\mathcal{B}} F) \simeq \text{hocolim}_{\mathcal{B}} (B \circ F)$ [23, Thm. 1.2] (statement as quoted in [22, Thm. 1.1]; textbook treatment Riehl [25, Ch. 5–6]); the proof below uses only its π_0 shadow (Lemma 10.3).

How the section populates the diagram. The transport data over $\mathcal{B}$ is supplied by the winding lift of the section's value-paths: a lift of a value-loop γ of A is a Γ with $\text{pr}_1 \circ \Gamma = \gamma \circ \exp$ (Theorem 6.6; [21, Def. 5.11, Prop. 4.2]); it is *unique* when the loop is tame — unique companion — ([21, Def. 4.7]) and *exists, and is itself a loop*, exactly when the signature criterion $\sigma \in \{0, -1\}$ holds on each obstruction interval ([21, Cor. 5.13]; the earlier citation to Cor. 5.22 is superseded by this pinned number). The residue- $\mathbb{C}$ zero-spheres of A appear in $\mathcal{T}_A$ as the *degenerate fibre* of this transport — an output of the construction, never an input — and the theorem computes the image of that fibre in $\pi_0(\mathcal{T}_A)$.

10 The concentricity theorem

Proposition 10.1 (Slice-regular Weierstrass factorization; the content of C3). *Let $A \in \mathcal{R}$ have a zero of order $m \geq 0$ at the origin and no non-real isolated zeros (for $A \in \mathcal{R}$ the latter is automatic: non-real zeros come in whole spheres, Lemma 10.2). Then on $\mathbb{O}^* \setminus \{N\}$ it factors as*

$$A = q^m R S h,$$

where q denotes the octonionic coordinate function (the identity section, so q^m carries the order- m zero at the origin), R is a slice-preserving factor vanishing exactly at the real zeros of A , S is a factor vanishing exactly at its spherical (residue- $\mathbb{C}$) zero-spheres, and h is never vanishing; each of R, S is the slice-regular Weierstrass product over the corresponding zeros. The factorization holds including the case where either family of zeros is infinite.

Proof. This is the factorization theorem of Altavilla–de Fabritiis [2, Prop. 3.1, Thm. 3.2], whose primary factors are the slice-regular ones of Gentili–Vignozzi [10]; convergence of the infinite products is part of the cited statement. The pair R, S is the residue split named in C3: R carries the residue- $\mathbb{R}$ (classically trivial) zeros and S the residue- $\mathbb{C}$ (classically nontrivial) ones, the latter infinite in number by C4. $\square$

Lemma 10.2 (Residue- $\mathbb{C}$ zeros are 6-sphere components of $\mathcal{H}_1$). *For $A \in \mathcal{R}$, the non-real zeros of A are 6-spheres, each a single connected component of the translation groupoid $\mathcal{H}_1$.*

Proof. A is slice-preserving, so on a sphere $\sigma + \gamma S^6$ ($\gamma > 0$) it has the form $A(\sigma + \gamma v) = \alpha(\sigma, \gamma) + v\beta(\sigma, \gamma)$ with α, β real (the real stem, [16]); since v is an imaginary unit, $\alpha + v\beta = 0$ forces $\alpha = \beta = 0$, independently of v . Hence a non-real zero occurs on a *whole* sphere $\sigma + \gamma S^6$ or at none of its points (the spherical-zero structure, [1]); real zeros are isolated real points (residue field $\mathbb{R}$). Each such sphere $\sigma + \gamma S^6$ is the G_2 -orbit of any of its points (Theorem 7.5: G_2 acts transitively on S^6 with stabiliser $SU(3)$), and the connected components of $\mathcal{H}_1$ are exactly the G_2 -orbits — in a groupoid two objects share a component iff a morphism joins them, and a morphism $x \rightarrow y$ of $\mathcal{H}_1$ is a $g \in G_2$ with $g \cdot x = y$. Therefore the residue- $\mathbb{C}$ zeros — the non-real ones — are exactly the 6-sphere components of $A^{-1}(0)$. $\square$

Lemma 10.3 (π_0 of a Grothendieck construction). *For a functor $F : \mathcal{B} \rightarrow \mathbf{Grpd}$, the connected-components functor carries the Grothendieck construction to the colimit of the component diagram:*

$$\pi_0\left(\int_{\mathcal{B}} F\right) \cong \operatorname{colim}_{\mathcal{B}} (\pi_0 \circ F).$$

Proof. The functor $\pi_0 : \mathbf{Cat} \rightarrow \mathbf{Set}$ is left adjoint to the inclusion of discrete categories, so it preserves colimits; and components of a category correspond one-to-one with components of its classifying space (Quillen [24, §1]). The identity is also direct at the level of categories. An object of $\int_{\mathcal{B}} F$ is a pair (b, x) ; a morphism (f, φ) projects to a morphism f of $\mathcal{B}$ and restricts, over a fixed base object, to a morphism of the fibre. Hence a zigzag joining (b, x) to (b', x') projects to a zigzag joining b to b' and, fibrewise, joins the band coordinates; two objects of $\int_{\mathcal{B}} F$ lie in one component exactly when their base objects lie in one component of $\mathcal{B}$ and their fibre coordinates are identified there, which is precisely $\operatorname{colim}_{\mathcal{B}} (\pi_0 \circ F)$. Passing to classifying spaces, this is the π_0 of Thomason's equivalence $B(\int_{\mathcal{B}} F) \simeq \operatorname{hocolim}_{\mathcal{B}} (B \circ F)$ [23] (see also [25, Ch. 5–6]). $\square$

Definition 10.4 (A -sections). An A -section is a section A of the ring $\mathcal{R}$ of slice-preserving slice-regular functions on $\mathbb{O}^* = S^8$ (Definition 5.1) — equivalently, a slice-preserving *semiregular* function on $\mathbb{O}^*$ (the meromorphic slice class; [7, §11]) — with the following four properties.

- (C1) *Meromorphic continuation; single simple pole.* A is meromorphically continued to $\mathbb{O}^*$ with exactly one pole; it is simple, lies at a real point, and its value there is the point at infinity $\infty = N$.
- (C2) *Infinite Euler product.* On a slice right half-space $\Omega_0 \subset \mathbb{O}^*$, $A = \exp(\sum_p \ell_p)$ for an *infinite* summable family $\{\ell_p\}$ of slice-preserving slice-regular functions each zero-free on Ω_0 ; in particular A is zero-free on Ω_0 .
- (C3) *Infinite Weierstrass factorization.* On $\mathbb{O}^* \setminus \{\text{pole}\}$,

$$A = q^m R e^g \prod_n \mathcal{E}(\cdot; q_n),$$

where q is the octonionic coordinate, $m \geq 0$, R is the slice-regular Weierstrass product over the residue- $\mathbb{R}$ (real) zeros of A , vanishing only on $\mathbb{R}$, g is slice-preserving entire, $\{q_n\}$ enumerates the residue- $\mathbb{C}$ zero-spheres of A , and $\mathcal{E}(\cdot; q_n)$ is the slice-regular Weierstrass primary factor of q_n (Proposition 10.1; [2, Prop. 3.1, Thm. 3.2], [10]); the factorization holds with either family infinite.

(C4) *Infinitely many residue- $\mathbb{C}$ zeros.* The index set $\{q_n\}$ is infinite. A closed zero of such a section is a real point (residue field $\mathbb{R}$) or a 6-sphere $S_{(\sigma, \gamma)}$ (residue field $\mathbb{C}$; [2]); residue- $\mathbb{C}$ names the classically nontrivial zeros (Part 2).

Theorem 10.5 (Concentricity). *Let A be an A -section (Definition 10.4). Then the residue- $\mathbb{C}$ zero spheres of A all lie in a single connected component of the total object $\mathcal{T}_A$ (Definition 9.9) — equivalently, they have one and the same image in $\pi_0(\mathcal{T}_A)$.*

Proof. Let $p : \mathcal{T}_A = \int_{\mathcal{B}} F \rightarrow \mathcal{B}$ be the Grothendieck construction of Definition 9.9. Hypotheses C1–C4 assemble the diagram; the residue- $\mathbb{C}$ zero-spheres then arrive as the degenerate fibre of its transport, an output; and π_0 reads off their component.¹

Assembly: the unique tame transport extending the exponential's base. The exponential's degenerate fibre over a real value is the level-indexed family of Lemma 6.3 — one real level, winding carried by the band — a structure of the space itself, absent over $\mathbb{C}$ ([17, Rem. 5.2(b)]; Definition 9.9). C2 and C3 extend the exponential structure over A : by C2, on Ω_0 the section is the exponential $A = \exp(\sum_p \ell_p)$ of an infinite summable slice-preserving family, zero-free there, so the continuation of the hypercomplex logarithm of A exists outright on Ω_0 (Theorem 6.6: a continuation is $\tilde{\gamma}$ with $\exp \circ \tilde{\gamma} = \gamma$); by C3, on $\mathbb{O}^* \setminus \{\text{pole}\}$ the section is the exponential expression $A = q^m Re^g \prod_n \mathcal{E}(\cdot; q_n)$ over its full divisor. On any slice $\mathbb{C}_I$ both are the one holomorphic stem A_I (Definition 9.4; Wang [16, Rem. 2.11]), agreeing on the overlap Ω_0 , hence coinciding wherever both are defined (Identity Theorem 3.3). The value-loops of A are therefore tame — unique companion ([21, Def. 4.7]) — so their lifts to the logarithm manifold, when they exist, are unique; and by C1 the section has exactly one pole, whose value is $\mathbf{n}$ (Remark 9.3), the cone through which the value-loops close: along these loops the signature criterion holds and the lift exists *and is itself a loop* ([21, Cor. 5.13]). This is the transport of Definition 9.9, canonically attached to A by C1–C4 jointly. Its degenerate fibre — the exp-preimage data over the real values met by the unique tame lift — is the level-indexed fibre of Lemma 6.3, and the residue- $\mathbb{C}$ zero-spheres $\{q_n\}$ of C3 are exactly this degenerate fibre, infinite by C4.

Through the commuting triangle $\pi \circ E = \exp$ ([17, Rem. 5.2(a)]), the unique tame lift traverses the logarithm manifold as a single closed loop ([21, Cor. 5.13]), and every point of the degenerate fibre it meets is, by Lemma 6.3, the level $\log r$ paired with an odd winding height $I(2k+1)\pi$: all multiplicity in the fibre lies in the winding direction ([21, Cor. 5.21]), none in the level. Since $\mathcal{B}$ is static — no morphisms between distinct levels (Definition 9.9) — the level is a conserved quantity along every zigzag of $\mathcal{T}_A$, and the degenerate fibre of the unique tame transport attached to the A -section — the residue- $\mathbb{C}$ zero-spheres $\{q_n\}$ of C3 — lies over a *single* level.

The cocartesian computation of π_0 . The fibre F is connected, so by Lemma 10.3

$$\pi_0(\mathcal{T}_A) \cong \operatorname{colim}_{\mathcal{B}}(\pi_0 \circ F) \cong \pi_0(\mathcal{B}),$$

under which a residue- $\mathbb{C}$ zero-sphere maps to the class of the base object beneath it, and two of them share a component of $\mathcal{T}_A$ if and only if they share that class. By the assembly and the placement above, the degenerate fibre lies over one and the same class — the zero-spheres over a

¹Remark 10.6 sketches an alternative, finality-based computation of π_0 ; it is expository and not used here.

single real level c , one for each winding index carried by the band — hence has a single image in $\pi_0(\mathcal{T}_A)$: one connected component.

Readback in $\mathcal{H}_1$. By Lemma 10.2 each residue- $\mathbb{C}$ zero-sphere is a G_2 -orbit component of $\mathcal{H}_1$, carried into the slice world by the section functor Φ (Theorem 9.7; Remark 9.8); distinct spheres are distinct orbits, hence pairwise disjoint, and by C4 there are infinitely many. The theorem states only the one-component conclusion; reading the component as a common *centre* is the downstream translation of Corollary 10.9. $\square$

Remark 10.6 (A second proof, via finality). The single-component conclusion also follows from Quillen’s §1 [24], on the same assembly, in either of two forms. *Finality:* by C1 the continuation of A carries every zero-bearing level of $\mathcal{B}$ to the north pole $\mathfrak{n}$, and the winding lift supplies a unique tame companion along each such continuation ([21, Def. 4.20, Cor. 5.13]); so the comma category from a zero-bearing level to $\mathfrak{n}$ is non-empty and connected, and $\mathfrak{n}$ is final over the zero-bearing part of $\mathcal{B}$ — the π_0 shadow of Theorem A (“if $Y \setminus f$ is contractible for every Y , then f is a homotopy equivalence”; connected and non-empty is the π_0 -level hypothesis; textbook form Riehl [25, §8.3, §8.5]). *Fibred form:* the projection $p : \int_{\mathcal{B}} F \rightarrow \mathcal{B}$ is precofibred with fibre the connected band, and Quillen’s corollary to Theorem A (pre(co)fibred with contractible fibres $\Rightarrow$ homotopy equivalence) has as its π_0 shadow exactly the bijection $\pi_0(\mathcal{T}_A) \cong \pi_0(\mathcal{B})$ used above. Thomason’s theorem [23] lifts the identity of Lemma 10.3 to the classifying-space equivalence $B(\int_{\mathcal{B}} F) \simeq \text{hocolim}_{\mathcal{B}}(B \circ F)$. The cocartesian argument above is the primary one; this is recorded as an alternative.

Theorem 10.7 (Connected is concentric: the dictionary). *Residue- $\mathbb{C}$ spheres of the degenerate family of Lemma 6.3 lie in one connected component of the static base — have one image in $\pi_0(\mathcal{B})$ — if and only if they share one real level, if and only if they are 6-spheres about one real centre: concentric.*

Proof. The components of the static base are the real levels by construction: the base is static, no morphisms between distinct levels, the level conserved along every zigzag (Definition 9.9); so one image in $\pi_0(\mathcal{B})$ is one level. By Lemma 6.3 the degenerate fibre over $-r$ is $\exp^{-1}(-r) = \{\log r + I(2k+1)\pi\}$ — one real level $\log r$, winding carried by the band — and its spheres have centre $\log r$ and radii $(2k+1)\pi$: one level is one centre. Spheres about one centre are concentric. No property of A -sections and no part of Theorem 10.5 enters. $\square$

Remark 10.8 (Dictionary vs. application). The dictionary above is logically independent of the collapse: its **uses** carries no arrow from Theorem 10.5, and the blueprint graph shows the two nodes unlinked. The application, Corollary 10.9, consumes both — the collapse supplies *one component*; the dictionary reads it as *one centre*.

Corollary 10.9 (Translation to the classical framework). *Identifying residue- $\mathbb{C}$ zeros with the classically nontrivial zeros, and residue- $\mathbb{R}$ zeros with the trivial ones (Part 2), such a section has its nontrivial zeros realised as infinitely many pairwise disjoint concentric 6-spheres about a single real centre.*

Proof. By Theorem 10.5 the residue- $\mathbb{C}$ zero-spheres lie in one connected component of $\mathcal{T}_A$. By the dictionary (Theorem 10.7) one component of the static base is one real level, hence one real centre c : the spheres are concentric about c . Their identification with the classical nontrivial zeros is Theorems 8.1 and 8.2; infinitude and pairwise disjointness are C4 and Lemma 10.2. $\square$

11 Corollaries

Corollary 11.1 ($\zeta_{\mathbb{O}^*}$ is an A -section). $\zeta_{\mathbb{O}^*}$ is an A -section (Definition 10.4): a section of $\mathcal{R}$ satisfying (C1)–(C4).

Proof. $\zeta_{\mathbb{O}^*} \in \mathcal{R}$ by slice-preservation (Theorem 7.2). The verification of (C1)–(C4) cites the classical facts of the Riemann zeta function (Part 1) and the slice-regular zero-geometry that translates them:

- **(C1) Meromorphic continuation; simple pole.** Classical fact: ζ is meromorphically continued with a single simple pole at $s = 1$ (Theorem 1.1); the value of $\zeta_{\mathbb{O}^*}$ there is the point at infinity, $\zeta_{\mathbb{O}^*}(1) = \infty = N$ (Definition 4.1).
- **(C2) Infinite Euler product.** Classical fact: $\zeta = \prod_p (1 - p^{-s})^{-1} = \exp(\sum_p \ell_p)$ on $\Re s > 1$, the family indexed by the infinitely many primes (Theorem 1.2).
- **(C3) Infinite Weierstrass factorization.** Supplied by the *slice-regular* Weierstrass factorization (Proposition 10.1; [2, Prop. 3.1, Thm. 3.2], [10]): $\zeta_{\mathbb{O}^*} = q^m \text{Re}^g \prod_n \mathcal{E}(\cdot; q_n)$, where $m = 0$ ($\zeta(0) = -\frac{1}{2} \neq 0$), R is the product over the residue- $\mathbb{R}$ zeros — the classically trivial zeros $-2, -4, \dots$, honest real zeros of $\zeta_{\mathbb{O}^*}$ — and $\{q_n\}$ enumerates its residue- $\mathbb{C}$ zero-spheres.
- **(C4) Infinitely many residue- $\mathbb{C}$ zeros.** Classical fact: ζ has infinitely many nontrivial zeros (Corollary 1.4; also Hardy, Theorem 1.5). By the residue dictionary below these are exactly the residue- $\mathbb{C}$ zeros, so $\{q_n\}$ is infinite.

The one remaining classical fact enters the downstream centre-pinning rather than (C1)–(C4): the *functional equation* $\xi(s) = \xi(1-s)$, $\xi(\bar{s}) = \overline{\xi(s)}$ (Theorem 1.1), used in Corollary 11.2. The *residue dictionary* — a closed zero of $\zeta_{\mathbb{O}^*}$ is a real point (residue field $\mathbb{R}$, the classically *trivial* zeros) or a 6-sphere $S_{(\sigma, \gamma)}$ (residue field $\mathbb{C}$, the classically *nontrivial* zeros) — is the slice-regular zero-geometry of Theorem 8.2 and [1]; it is what makes “residue- $\mathbb{C}$ ” and “nontrivial” the same notion. Thus (C1)–(C4) hold. $\square$

Corollary 11.2 (Riemann Hypothesis). *Every nontrivial zero of the classical Riemann zeta function has real part $\frac{1}{2}$; in particular infinitely many zeros lie on the line $\Re s = \frac{1}{2}$.*

Proof. By Corollary 11.1, $\zeta_{\mathbb{O}^*}$ satisfies (C1)–(C4); by Theorem 10.5 the residue- $\mathbb{C}$ zero-spheres of $\zeta_{\mathbb{O}^*}$ lie in a single connected component of $\mathcal{T}_{\zeta_{\mathbb{O}^*}}$. By Corollary 10.9 that one component is a single real centre c , so the residue- $\mathbb{C}$ zeros are infinitely many concentric 6-spheres about c ; under the residue dictionary (Corollary 11.1) these are the classically nontrivial zeros, so every nontrivial zero has real part c . The functional equation $\xi(s) = \xi(1-s)$ (Theorem 1.1) sends a zero of real part c to one of real part $1-c$; applied to the *common* centre it forces $c = 1-c$, whence $c = \frac{1}{2}$ (this is Theorem 8.3 read on $\zeta_{\mathbb{O}^*}$). Therefore every nontrivial zero has real part $\frac{1}{2}$, and by (C4) infinitely many lie on $\Re s = \frac{1}{2}$. $\square$

Remark 11.3 (Status and formalization). Theorem 10.5 rests on the cited slice-regular inputs — the Weierstrass factorization (Proposition 10.1; [2, Prop. 3.1, Thm. 3.2], [10]), the winding lift and its criteria (Theorem 6.6, Proposition 6.7; [21, Def. 5.11, Def. 4.20, Cor. 5.13, Cor. 5.21]), and the exponential’s degenerate fibre (Theorem 6.1; Lemma 6.3) — together with the identity $\pi_0(\mathcal{T}_A) \cong \text{colim}_{\mathcal{B}}(\pi_0 \circ F)$ of Lemma 10.3. A second proof from Quillen’s §1 (Theorem A and its pre(co)fibred corollary [24]; Riehl [25, §8.3]) is recorded in Remark 10.6. The placement sentence remains the author’s and is conspicuously marked inside the proof of Theorem 10.5; the static-base reading is fixed, so the sentence must speak of levels and winding only.

The translation from “one connected component” to “one real centre,” and thence to concentric 6-spheres and the classical nontrivial zeros, is Corollary 10.9: the components of the static base are the real levels (Definition 9.9), and the equivalence Theorems 8.1 and 8.2 perform the identification. It is a corollary of the theorem, kept logically separate: the Concentricity Theorem itself states and uses only the one-component conclusion.

A machine formalization is future work; nothing in this document has been run in Lean. The categorical objects rest on existing Mathlib foundations — the translation groupoid $\mathcal{H}_1$ (`CategoryTheory.ActionCate`), the Grothendieck construction $\mathcal{T}_A$ (`CategoryTheory.Grothendieck`), and π_0 (`CategoryTheory.ConnectedCompon`) — while the octonionic and G_2 data and the band $U(1)$, absent from Mathlib, enter as axioms carrying their citations [19]. The section functor Φ (Theorem 9.7) and the winding lift (Theorem 6.6) are the constructions to formalize; the one-component conclusion is then the π_0 computation of Lemma 10.3.

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